

Module 14: Macroevolution — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Define a "species" using the Biological Species Concept.
2. Distinguish between prezygotic and postzygotic reproductive barriers.
3. Classify examples of isolation mechanisms (e.g., behavioral, temporal).
4. Differentiate between allopatric and sympatric speciation.

The Big Picture

Microevolution is the changing of allele frequencies in a single gene pool.

Macroevolution is simply microevolution scaled up over millions of years, resulting in the splitting of one gene pool into two distinct ones: the formation of new species. To get a new species, you have to stop gene flow (reproduction) between two groups long enough for them to evolve far apart.

Key Terms & Concepts

1. The Biological Species Concept

How do we decide if two animals are the same species? Do they just look alike? The most widely used definition is the **Biological Species Concept**. - Two organisms belong to the same species if they can **interbreed in nature and produce viable, fertile offspring**. - *Example*: A horse and a donkey can interbreed (making a mule), but mules are sterile (they cannot have babies). Therefore, horses and donkeys remain separate species. - *Limitation*: We cannot use this concept for extinct fossils (we don't know who successfully mated) or asexual organisms (like bacteria).

2. Reproductive Barriers

For speciation to occur, gene flow must stop. Mother Nature enforces this via reproductive barriers. They are split into two categories based on a simple dividing line: **the formation of a fertilized egg (a zygote)**.

Prezygotic Barriers (Before the Zygote Forms)

These prevent mating or fertilization from ever happening. - **Habitat Isolation:** They live in the same region but different habitats (e.g., one snake lives in the water, a closely related snake lives on land). - **Temporal Isolation:** They breed at different times (e.g., one skunk breeds in spring, another in fall). - **Behavioral Isolation:** They have different courtship rituals (e.g., female fireflies only respond to the flash pattern of their own species). - **Mechanical Isolation:** Physical incompatibility. (Their parts do not fit together). - **Gametic Isolation:** The sperm of one species cannot physically fertilize the egg of the other.

Postzygotic Barriers (After the Zygote Forms)

Mating happens and an egg is fertilized, but the offspring fails to thrive. - **Reduced Hybrid Viability:** The hybrid offspring is frail and dies before reaching adulthood. - **Reduced Hybrid Fertility:** The hybrid survives and is strong, but is entirely sterile (like a mule). - **Hybrid Breakdown:** The first generation of hybrids is fine and fertile, but when those hybrids mate, their offspring are weak and sterile.

3. Mechanisms of Speciation

How does one species physically split into two? - **Allopatric Speciation** ("Allo" = other, "patric" = country): A physical geographic barrier separates a population into two. (e.g., A river reroutes and splits a forest, separating a population of squirrels who evolve apart over millions of years until they can no longer interbreed). - **Sympatric Speciation** ("Sym" = same, "patric" = country): Speciation occurs in populations that live in the exact same geographic area. This happens via behavioral divergence, sexual selection, or (especially in plants) massive sudden mutations like polyploidy (having extra sets of chromosomes).

Study Tips

1. **Focus on the timeline of the barriers.** Remember the word "zygote" means the fertilized egg. Ask yourself: "Did a sperm reach the egg?" If no, it's prezygotic. If yes but the offspring has a problem, it's postzygotic.
2. **Evolution is ongoing.** Species are not perfectly static categories. Speciation is a moving slider. You might find groups of organisms actively in the middle of splitting apart today!